Assessing Trends in Wildfire Severity in Catalonia Over the Last Decade: A Multi-Class Classification Approach Using U-Net

https://github.com/pablo-fdz/DL_improc_final_project

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Research question: Has wildfire severity increased in Catalonia during the last decade? By training a multi-class classification convolutional neural network, we assess whether there has been an increase in the magnitude of the damage caused by wildfires in the last 10 years.

01 Motivation

According to the sixth IPCC report, the **frequency and extent of wildfires across the globe is increasing** with climate change, primarily driven by an increase in the average temperature (Parmesan et al., 2022). In Europe, the Southern region is expected to be one of the main victims of this trend, with a yearly GDP loss linked to wildfires of 13-21 billion euros (Meier et al., 2023) and a progressive deterioration of the region's ecosystem.

This research project estimates **whether there has been an increasing trend in the severity** of wildfires in Catalonia in the period 2014-2023, which could provide valuable information for prevention tasks and climate change adaptation strategies.

02 Background

Much of the work in this field has been devoted to the **delineation of burned areas** (Farasin et al., 2020; Knopp et al, 2020) rather than to **severity estimation of areas affected by wildfires** (the former is a binary classification problem subsumed in the latter, which is a multi-class classification or regression problem).

While there has been some research in the wildfire severity assessment using satellite imagery (Nancy et al., 2008; Rosalie and Lynham, 2008), imagery technology and machine/deep learning knowledge has considerably improved in recent years. Furthermore, the cited sources estimate wildfire severity in a significantly different ecosystem than the one that is object from this study (boreal forests vs. mediterranean forests).

The authors from the labelled dataset that is used in this project (see the "Data" section) test the performance of the UNet architecture with their data (which predominantly contains satellite imagery from wildfires in Southern Europe), reaching an average accuracy of 94%, a precision of 74% and a recall of up to 93% for **binary classification** (Colomba et al., 2022).

03 Data

1. Catalonia Fire Polygons from 2014-2023

Fire polygon data for Catalonia from 2014-2023 obtained from:

- **Source**: Generalitat de Catalunya - Departament d'Agricultura, Ramaderia, Pesca i Alimentació
- **URL:** https://agricultura.gencat.cat/ca/serveis/cartografia-sig/bases-cartografiques/boscos/incendis-forestals/incendis-forestals-format-shp/
- **Format:** Shapefiles containing fire boundaries, activation dates, and municipality information.
- **Usage**: the activation date and the coordinates of the bounding box of each of the fire polygons are used for retrieving Sentinel-1 and Sentinel-2 images from those wildfires.
- **Figures**: Figure 1: Wildfire Frequency Registered in Catalonia in 2014-2025.

2. Labelled Dataset for Training

Annotated dataset of 73 wildfires in Europe in the period 2017-2019:

- **Source**: Colomba et al. (2022)
- **URL:** https://zenodo.org/records/6597139
- **Format:**
  - Pre-fire (1 month before activation date) and post-fire (1 month after activation date) TIFF and PNG images from Sentinel-1 GRD and Sentinel-2 L2A of 73 wildfires in Europe from 2017 to 2019. Sentinel-1 images are in IW mode and with VV+VH polarization, with 4 bands (VV, 2VH, |VV|/|VH|/100 and coverage mask). Sentinel-2 images come with 13 bands (B01-B12 plus the coverage mask).
  - Pre-fire and post-fire PNG images of the coverage mask.
  - Post-fire TIFF and PNG images of the fire severity mask, with 5 classes: 0 (unburned) and 1-4, from low to complete wildfire damage severity.
- **Usage**:
  - Tiled the images into uniform 256x256 patches, ensuring complete coverage of the original image (but with overlap).
  - Trained a UNet-based multi-class classifier on the fire severity masks and Sentinel-1 and Sentinel-2 post-fire images (with all of the channels) and evaluated performance with the test split.
- **Figures**:
  - Figure 2: Sample Sentinel-2 image, coverage and fire masks with overlayed and reconstructed tiles.
  - Figure 3: Class imbalance.
  - Figure

3. Inference Dataset: Sentinel-1 and Sentinel-2 images of Catalonia

Out of 211 registered wildfires in Catalonia in the period 2014-2023, we retrieve 200 Sentinel-1 and/or Sentinel-2 images of each wildfire (95% of the total).

- **Source**: authors, using the Catalogue and Processing APIs from Sentinel Hub.
- **Format:** Post-fire (1 month after activation date) TIFF images from Sentinel-1 GRD and Sentinel-2 L2A, in the same bands as the labelled dataset, with the bounding box of the wildfire polygon at the center of each image. Minimum dimension of 512 pixels. The Sentinel-2 L2A images retrieved for each wildfire were the earliest images with less than 10% cloud coverage.
- **Usage**:
  - Same pre-processing as labelled dataset (tiling).
  - After having trained a UNet-based model, images are used for inference.
- **Figures**: Figure 5. Polygon, bounding box and buffer and the retrieved Sentinel-2 image for the fire in Òdena in the 26th July 2015.

04 Methodology

1. Pre-processing

Both annotated and inference data is **tiled** into uniform 256x256 patches. For the labelled data, tiles containing **no coverage** from either Sentinel-1 or Sentinel-2 are **removed**, leaving 3230 and 3176 tiles for training-validation-test with Sentinel-2 and Sentinel-1 images, respectively. All of the channels for Sentinel-1 (4 bands) and Sentinel-2 (13 bands) are kept.

2. Training

We train two different **UNet-based convolutional networks** on **multi-class classification**, one with the Sentinel-1 tiles and another with the Sentinel-2 tiles. We use a weighted cross-entropy loss (due to the existence of class imbalance), Adam optimizer and adapt the learning rate, weight decay, number of epochs and patience to the observed patterns when training with Sentinel-1 and Sentinel-2 images. Figures 6 and 7 show the evolution of training and validation loss and the intersection over union metric for the training with Sentinel-1 and Sentinel-2 images, respectively.

3. Evaluation on the test set

We extensively evaluate the classification performance of both of our trained models on the test set, both numerically (with classification metrics adapted to multi-class tasks) and visually. The Sentinel-2 model far outperforms the Sentinel-1 model with obtained accuracy, precision, recall and F1-Score of 72% , 86%, 72%, and 76% respecively (weighted averages) and 23%, 80%, 23%, and 27%, respectively. We can attribute this to a) a higher availability of data (the 9 extra color bands provide an extra 225% of data) and b) the significance of the information carried by the B01-B12 color bands.

4. Inference with Catalonia's wildfires satellite imagery

Given that the best-performing model observed is the one trained on Sentinel-2 images, we use the trained model to classify the images at the pixel level and to answer two different questions:

1. What has been the trend in the **burnt area in the last decade** (binary classification)?
2. Of the total area burned each year, **has severity increased**?

05 Results and Conclusions

We have classified, at the pixel level, whether a certain area has been burned and, if so, which is the associated severity. These are the results:

1. We cannot statistically conclude that the burnt area in Catalonia has followed an increasing trend in the last decade

Our model seems to adjust to the trend of wildfire frequency registered in Catalonia in 2014-2023 (Figure 1), as shown in Figure 10. Nevertheless, **we cannot reject the null hypothesis that there is a constant trend in the burnt area from wildfires in Catalonia.**

2. Wildfires have not (statistically) increased in severity in the last decade

Through the coefficients of a simple linear regression, **we cannot reject the null hypothesis that the fire severity of wildfires in Catalonia has kept constant during this last decade**. Visually, Figures 11 and 12 confirm these insights.

3. On the models and potential for inference through UNet or similar models

The Sentinel-2 model performance, despite the limited training process, suggests a potential for U-Net models in the assessment of fire-related damage when provided with data from the full Sentinel-2 spectral range. On the other hand, we do not foresee such potential with Sentinel-1 data.

By looking closely at the images, we find that the model struggles with the different landscapes in the dataset. This is shown, for instance, in its tendency to predict differently-looking areas of the image in different classes in tiles with only one true label. For future research, we suggest the training of specific models for the different climate regions as to allow them to further learn and specialize in the distinction of different soil types. This could also be achieved, partly, by a larger dataset and a much longer traingng process if better computational resources are available.

Figures - Data Preparation and Exploration

Figure 1. Wildfire Frequency Registered in Catalonia in 2014-2023

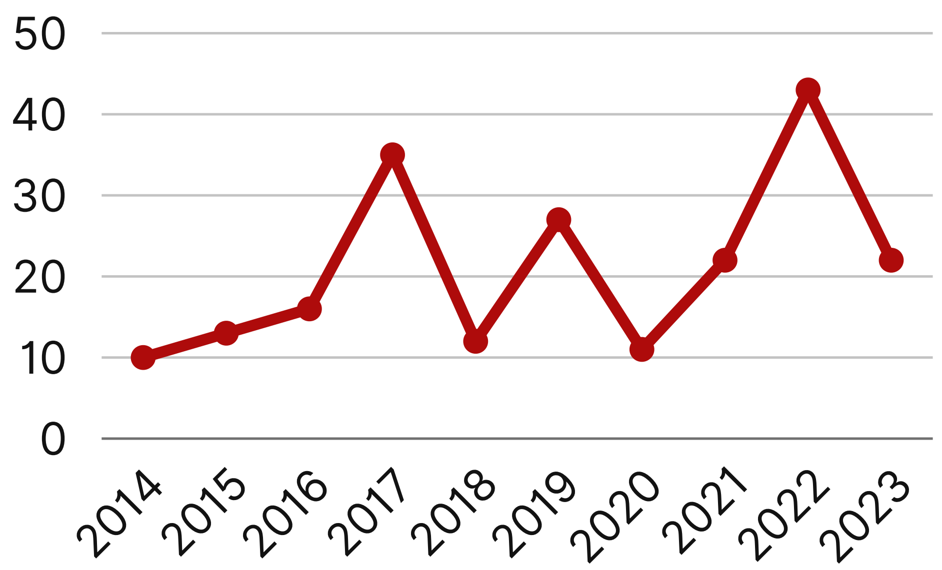


Figure 2. Sample Sentinel-2 image, coverage and fire masks with overlayed and reconstructed tiles

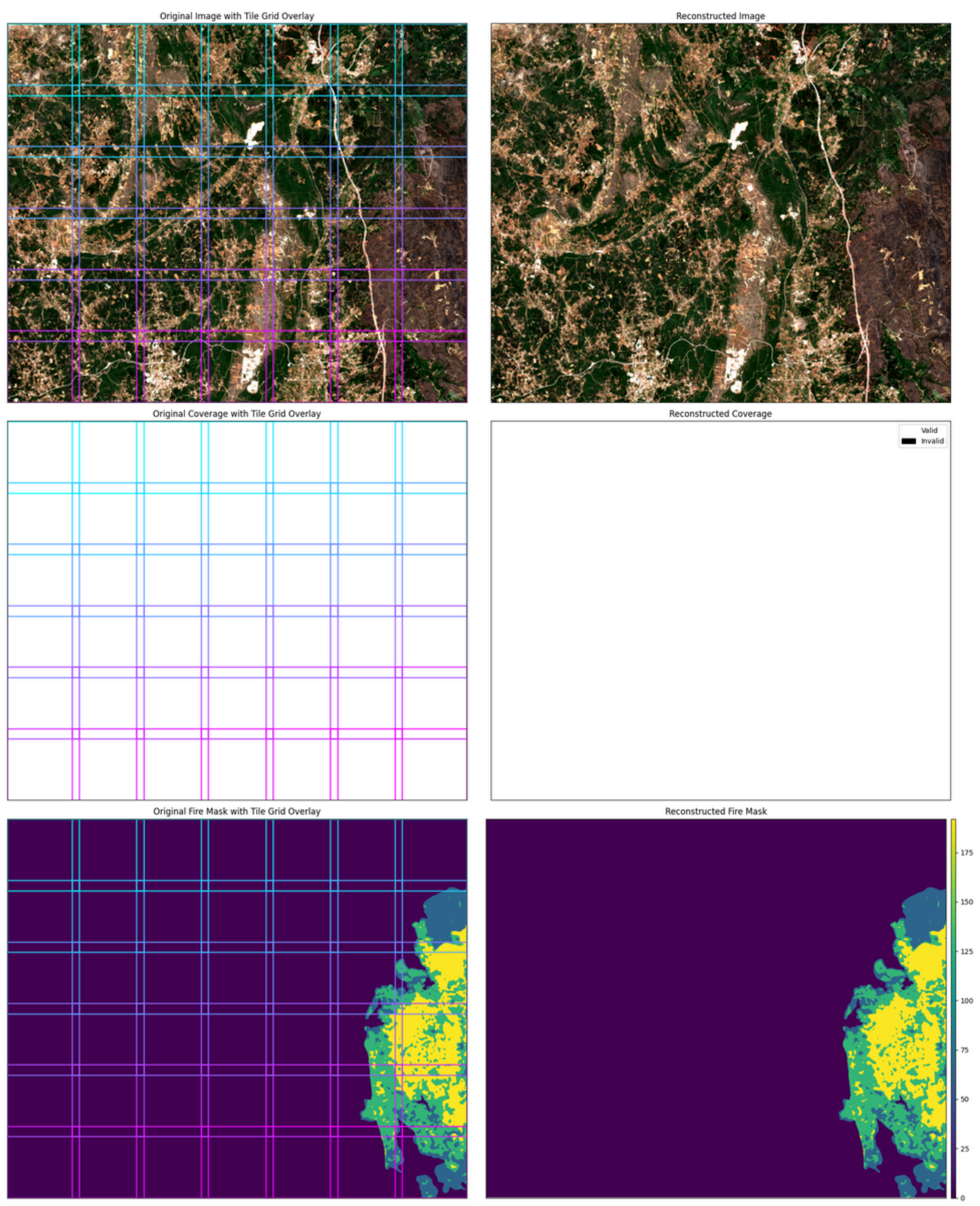


Figure 3. Class Balance

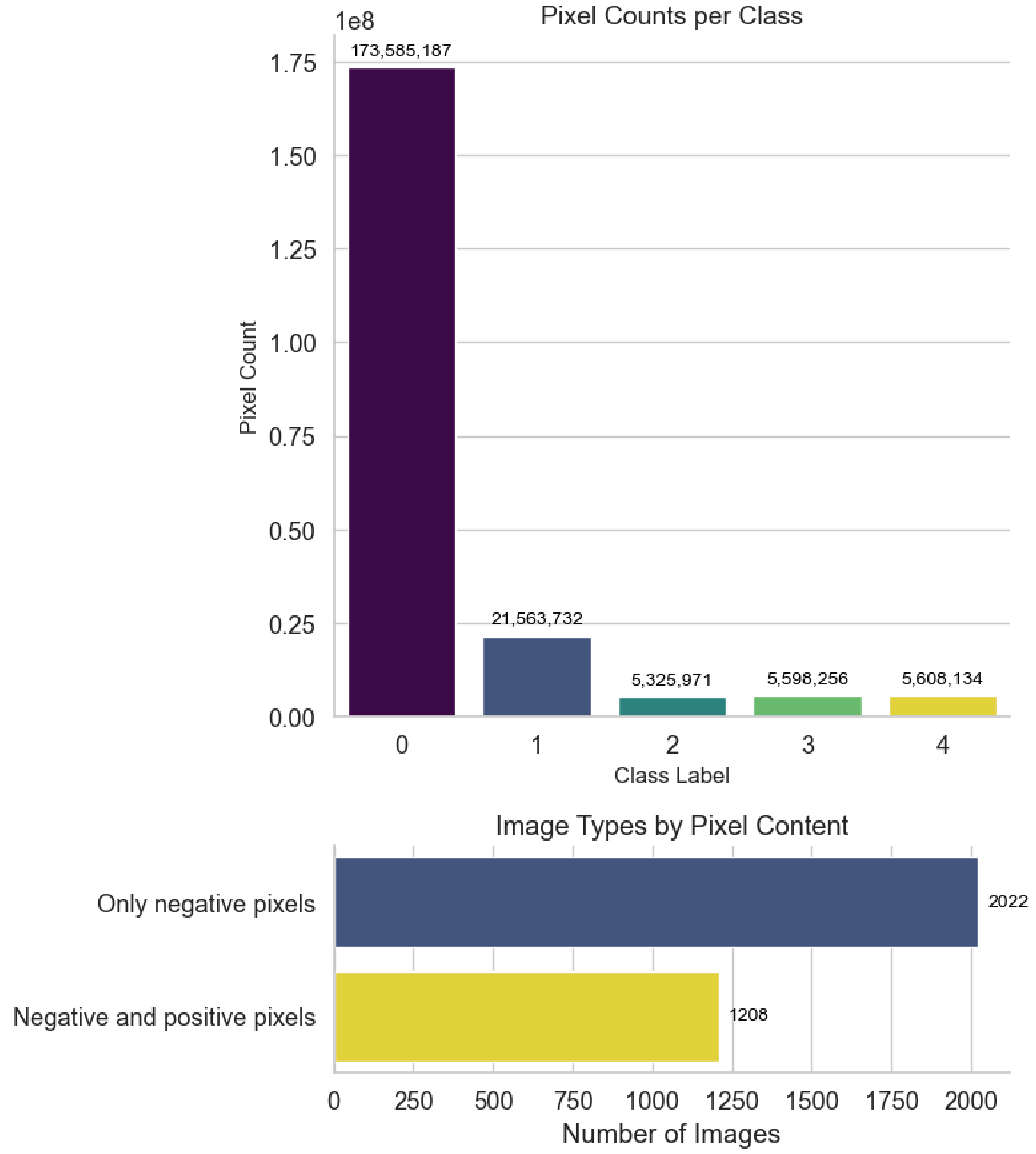


Figure 4. Visualisation vegetation-state related indices (NBR, NDVI, RE-NDVI, NDMI)

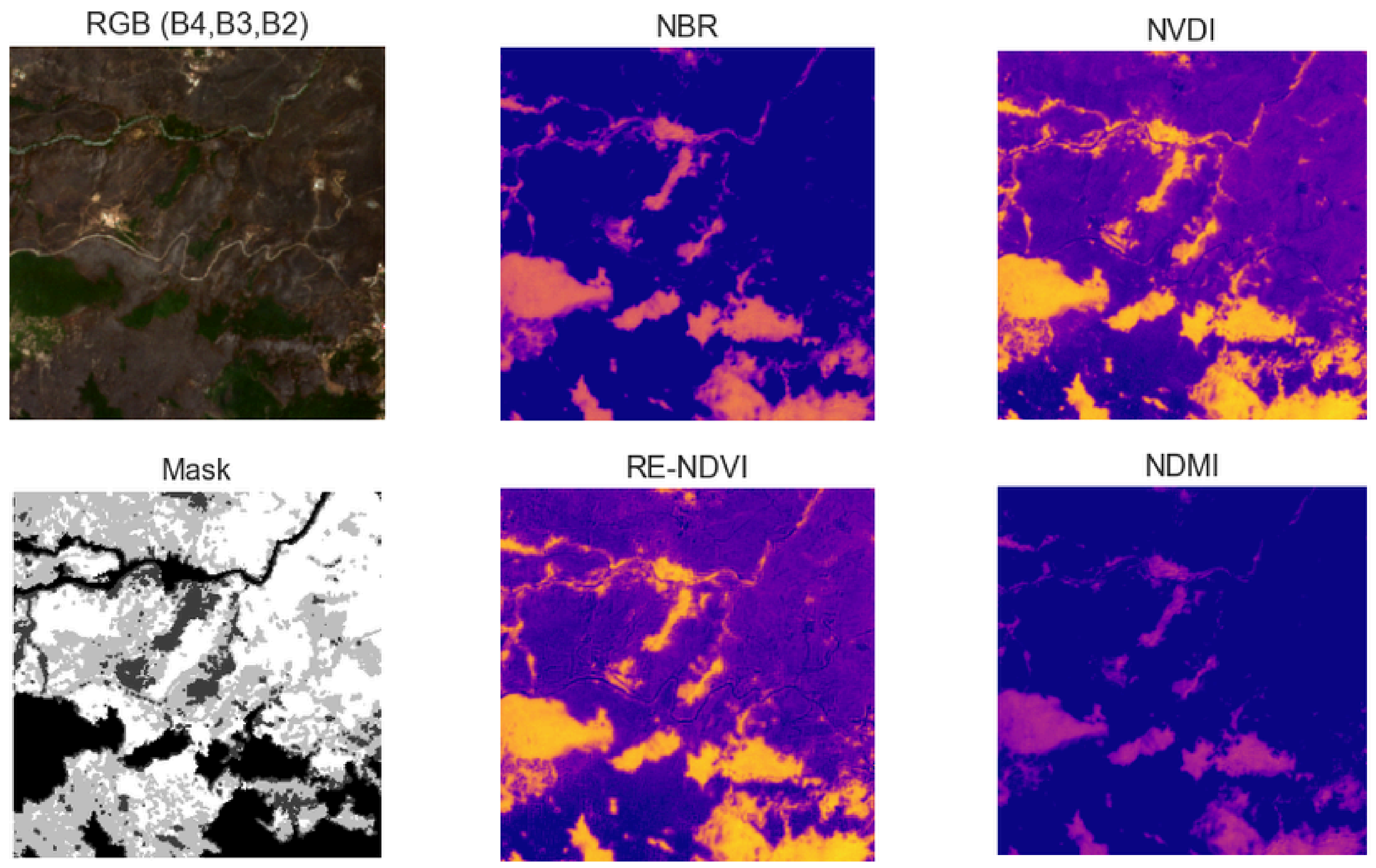
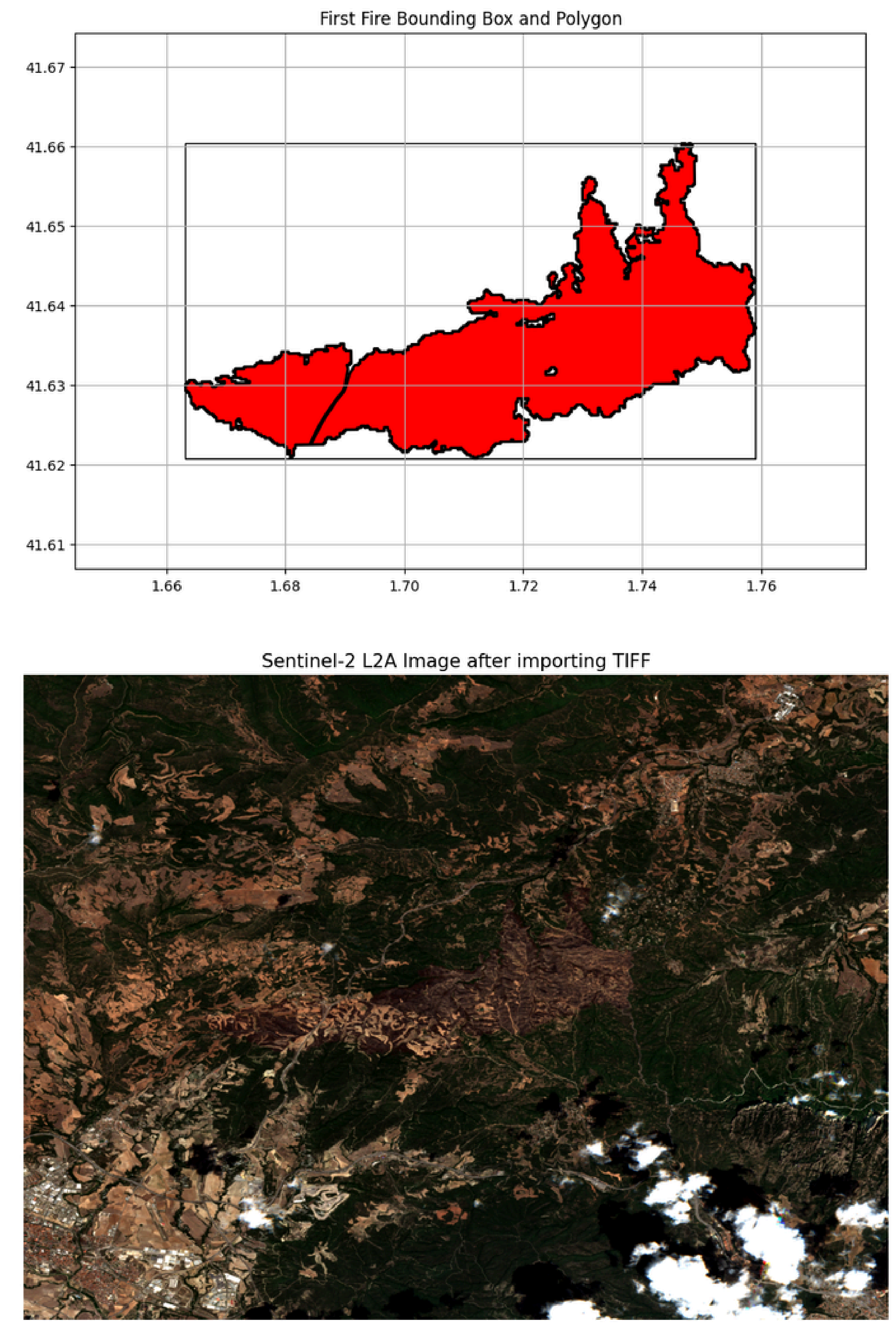


Figure 5. Polygon, bounding box and buffer and the retrieved Sentinel-2 image for the fire in Òdena in the 26th July 2015



Figures - Model Training

Figure 6. Loss and IoU over epochs with model trained on Sentinel-1 images

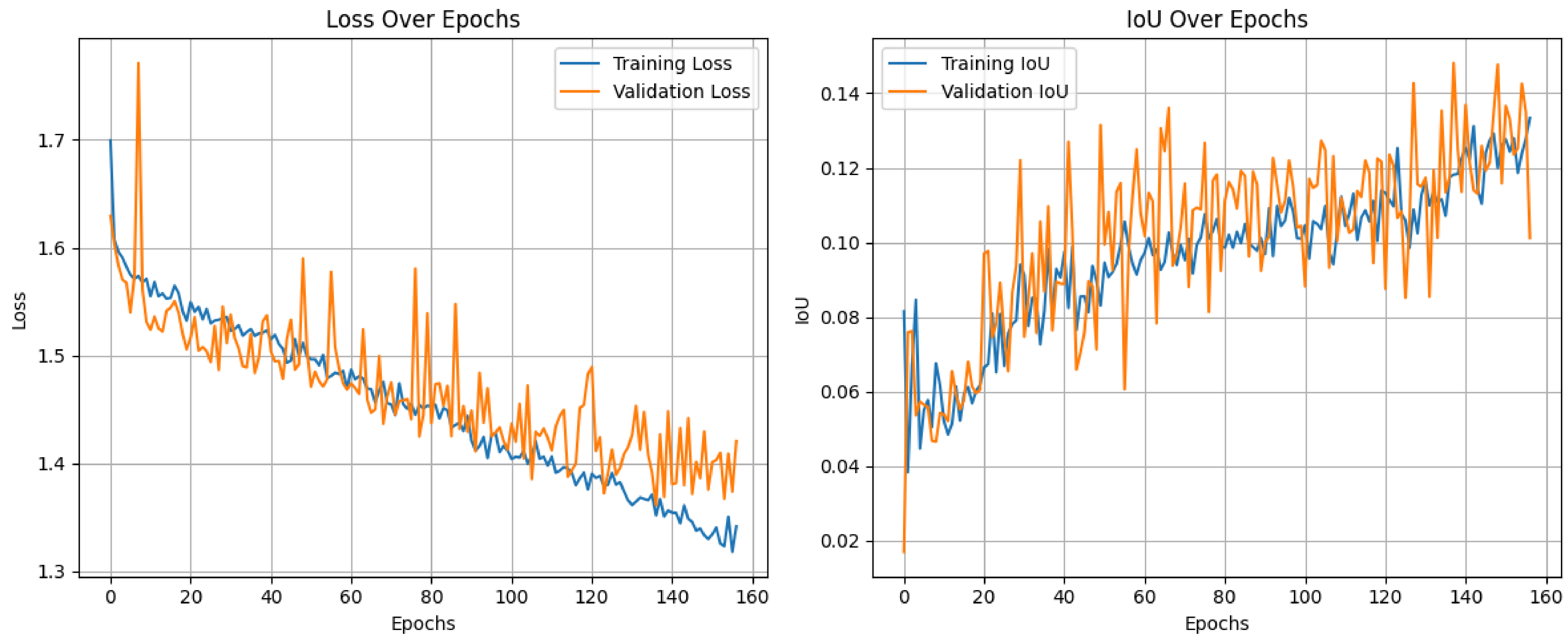
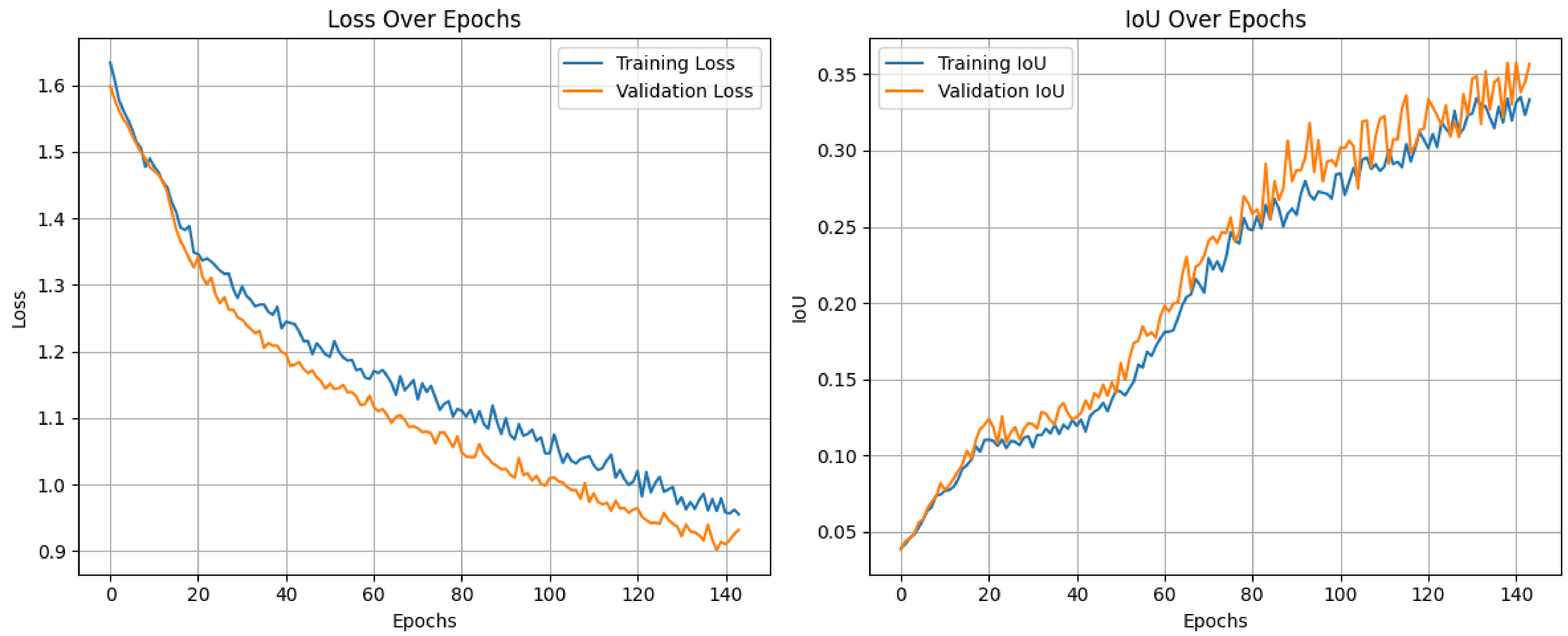
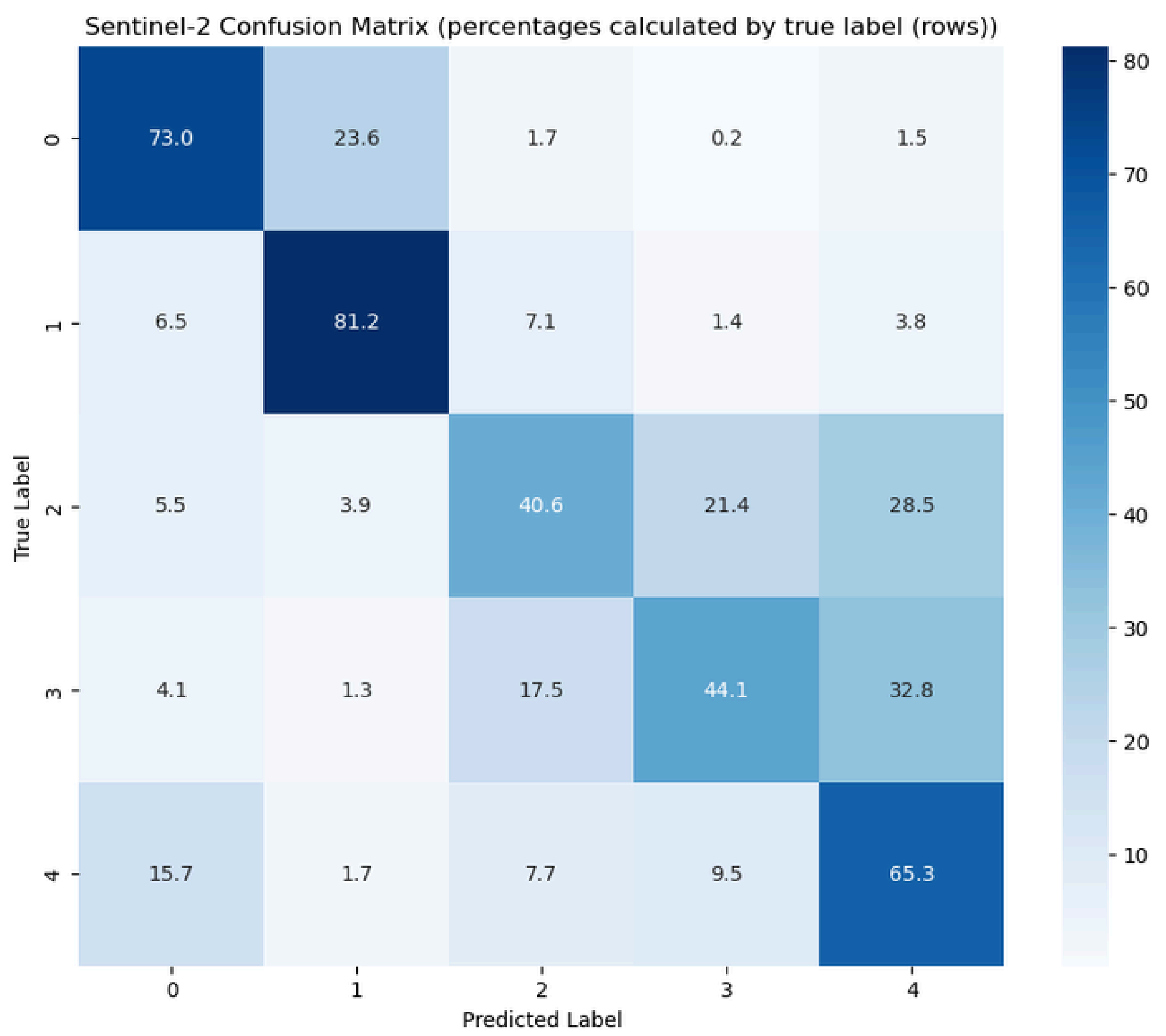


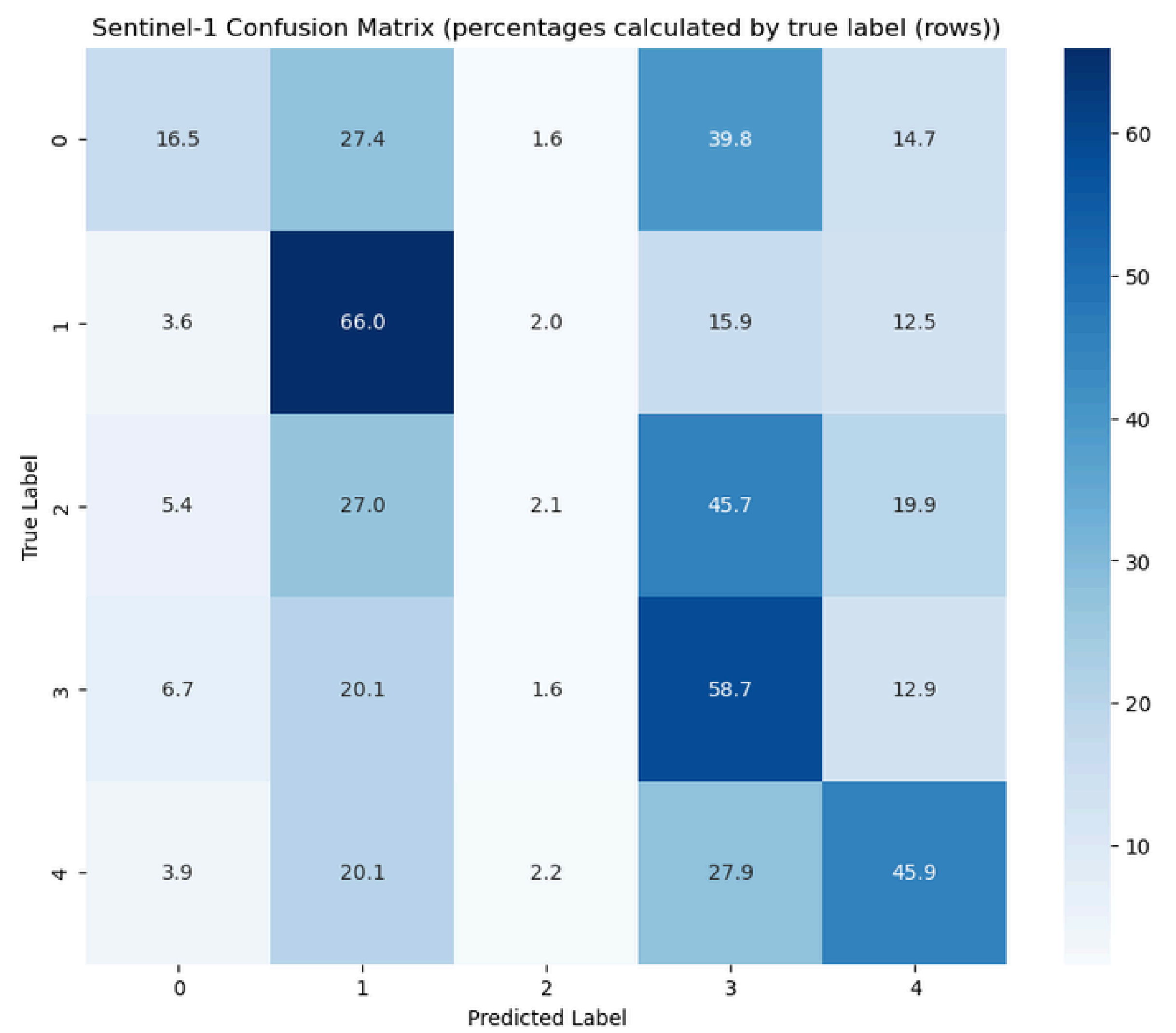
Figure 7. Loss and IoU over epochs with model trained on Sentinel-2 images



Figures - Model Performance

Figure 8. Confusion matrices



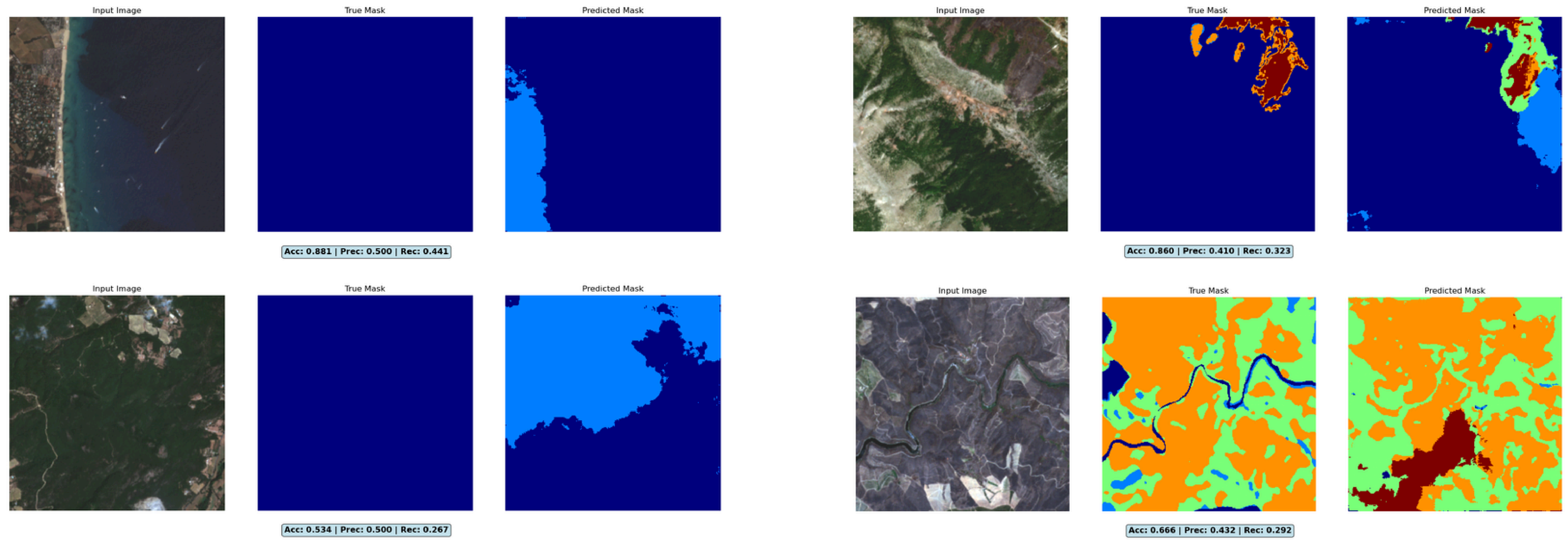


Sentinel-2: Most predicted label is the true label for all classes while the 2nd-most predicted label is one of the adjacent labels for all classes except #4, for which, surprisingly, it's class 0.

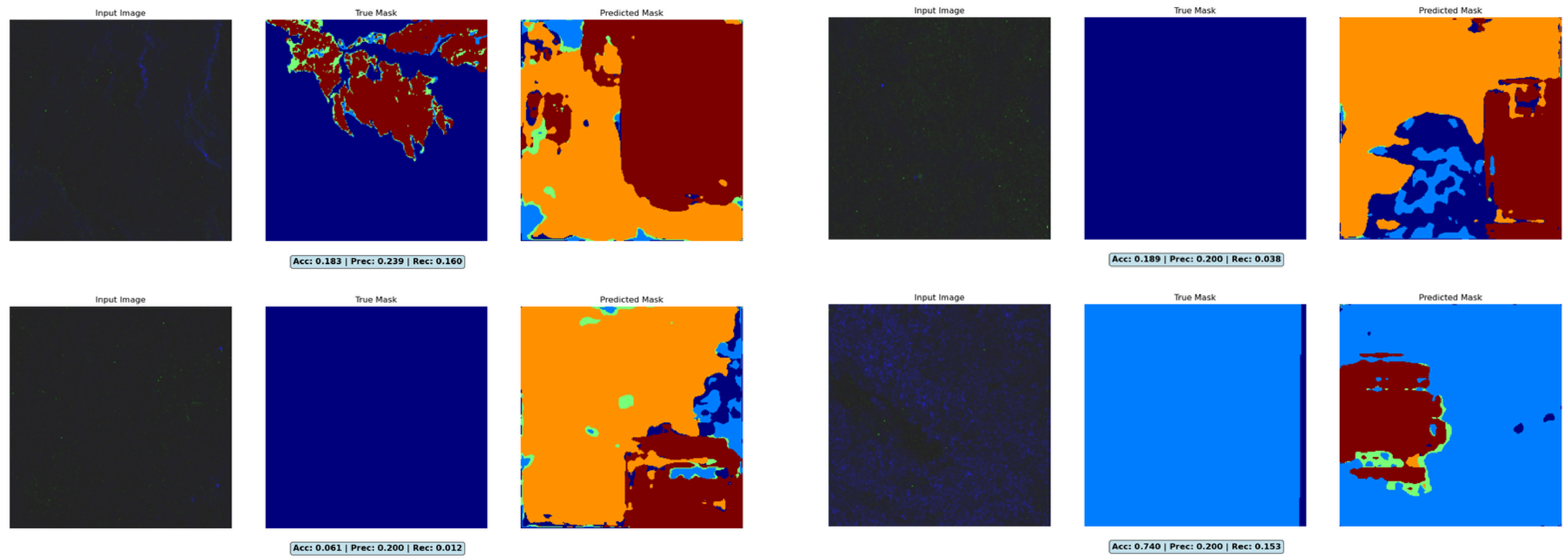
Sentinel-1: For 3 out of the 5 classes the most predicted label is the correct one, but it is much harder to determine a pattern. There is a big tendency towards predicting classes 1 and 3, despite that not being coherent with the balance of the classes in the dataset.

Figure 9. Samples of satellite image, true mask, and predicted mask

SENTINEL-2



SENTINEL-1



Color legend (class in parenthesis): Dark blue (0), Light blue (1), Bright Green (2), Yellow/Orange (3), Dark red (4)

Figures - Inference

Figure 10. Estimated Annual Burnt Area in Catalonia (2015-2023)

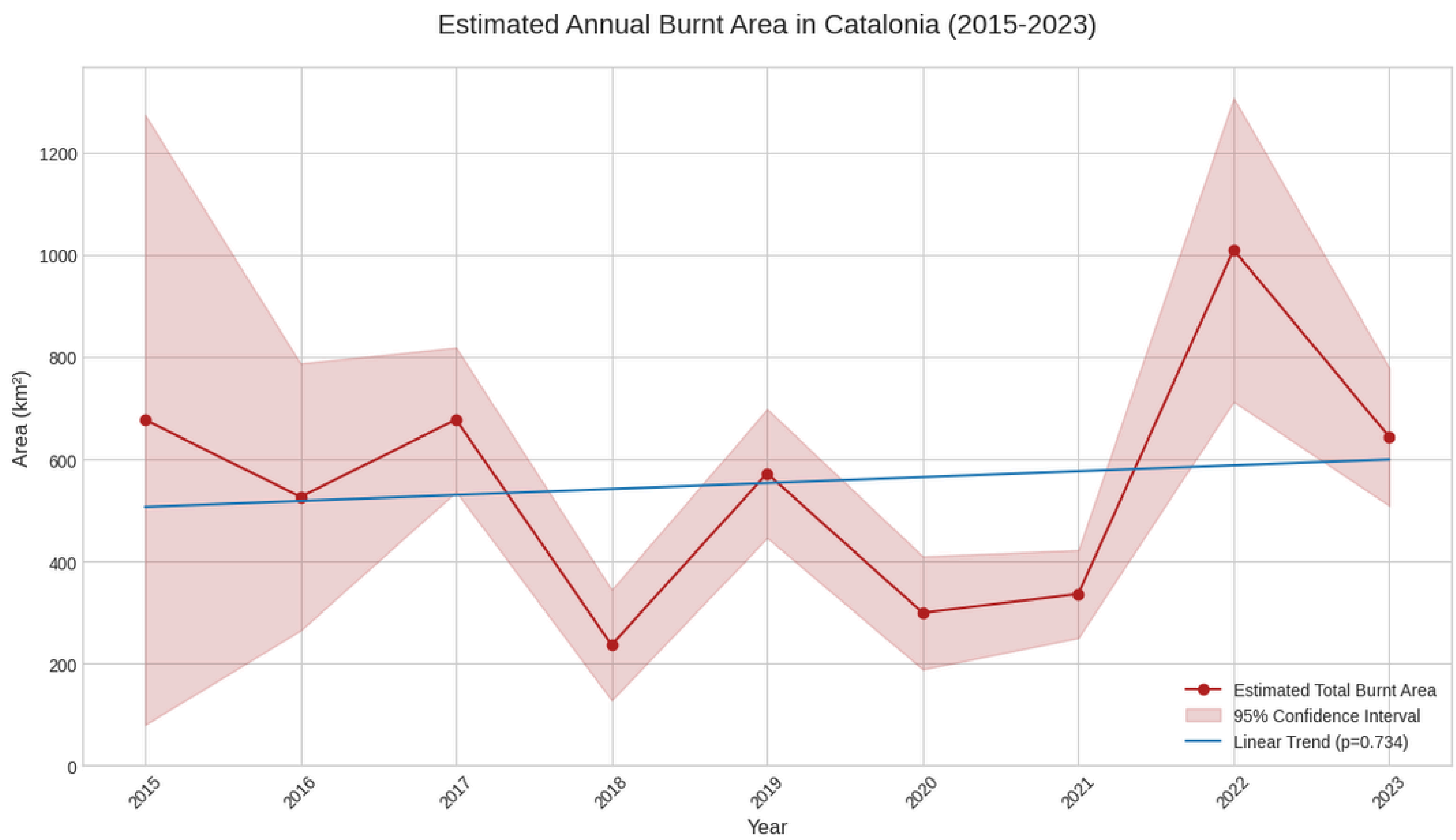


Figure 11. Average Wildfire Severity in Catalonia (2015-2023)

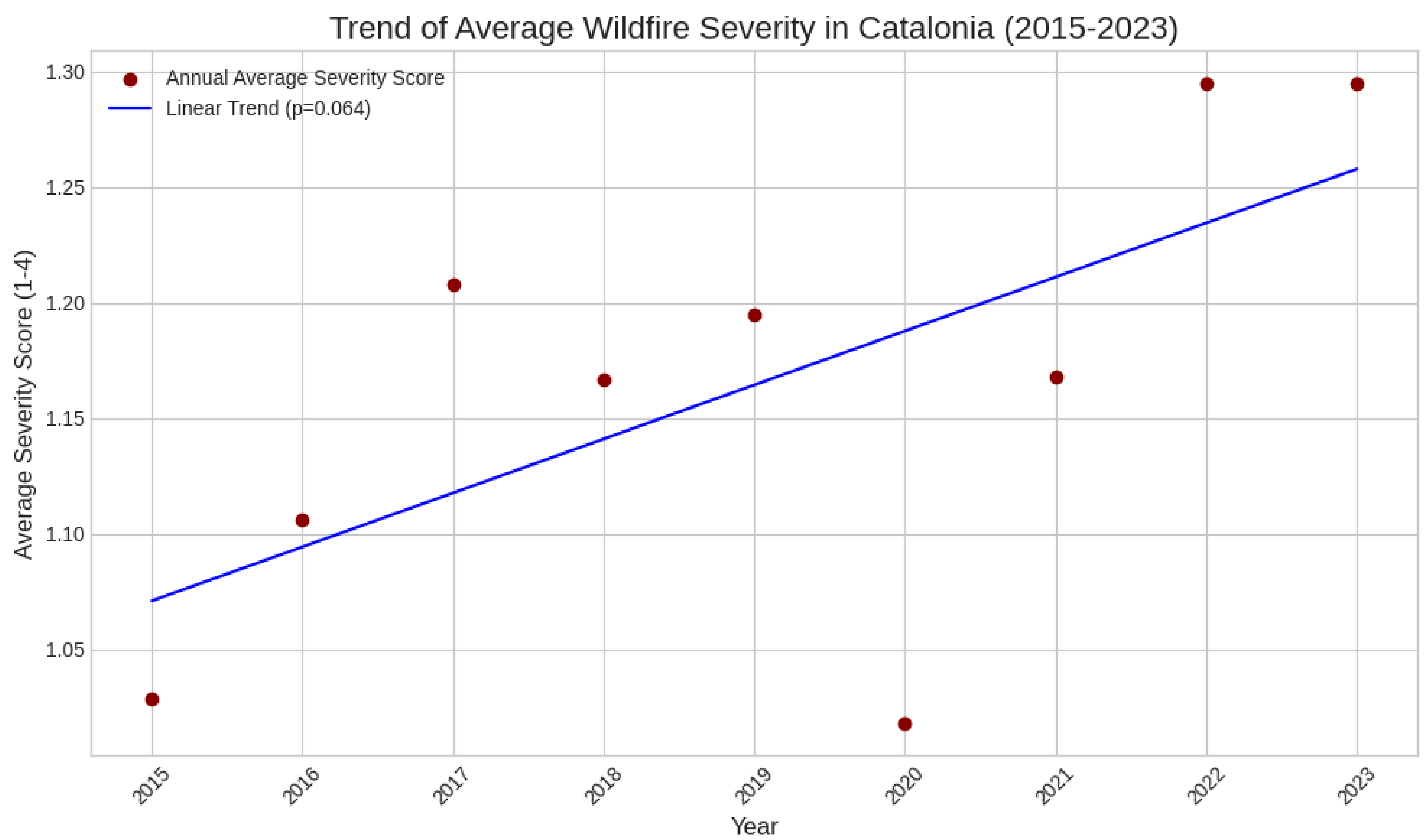
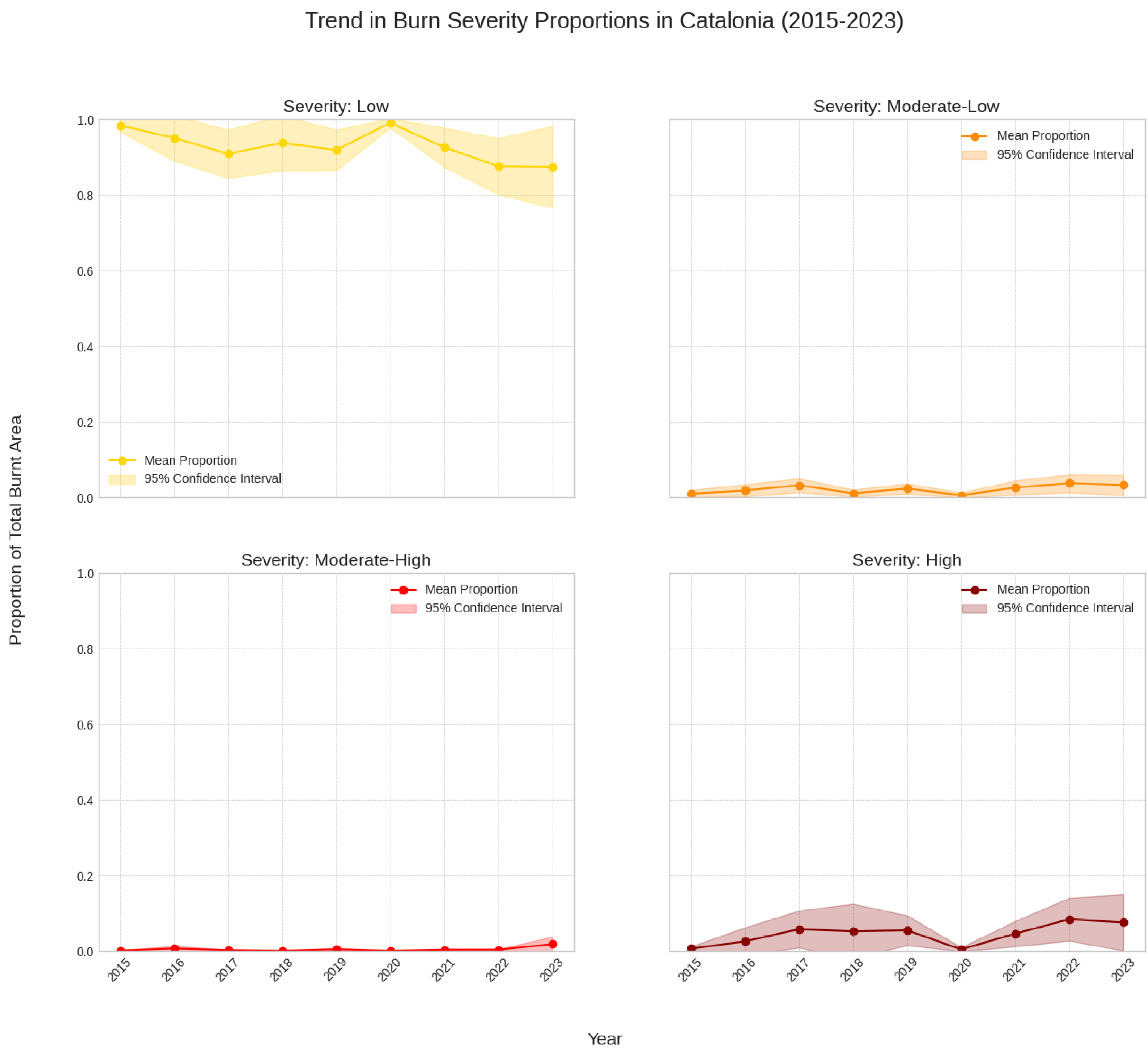


Figure 12. Trend in Burn Severity from wildfires in Catalonia (2015-2023)



*Area estimates are generated considering the Sentinel-2 resolution.

References

- Colomba, L., Farasin, A., Monaco, S., Greco, S., Garza, P., Apiletti, D., Baralis, E., & Cerquitelli, T. (2022). A dataset for burned area delineation and severity estimation from satellite imagery. Proceedings of the 31st ACM International Conference on Information & Knowledge Management.
- Farasin, A., Colomba, L., Palomba, G., Nini, G., & Rossi, C. (2020). Supervised Burned Areas delineation by means of Sentinel-2 imagery and Convolutional Neural Networks. In Proceedings of the 17th International Conference on Information Systems for Crisis Response and Management (ISCRAM 2020) (pp. 24–27).
- Knopp, L., Wieland, M., Rättich, M., & Martinis, S. (2020). A deep learning approach for burned area segmentation with Sentinel-2 data. Remote Sensing, 12(15), 2422. https://doi.org/10.3390/rs12152422
- Meier, S., Elliott, R. J. R., & Strobl, E. (2023). The regional economic impact of wildfires: Evidence from Southern Europe. Journal of Environmental Economics and Management, 118(102787), 102787. https://doi.org/10.1016/j.jeem.2023.102787
- Nancy, H. F., French, E. S., Kasischke, R. J., Hall, K. A., Murphy, D. L., Verbyla, E. E., & Hoy, J. L. (2008). Using Landsat data to assess fire and burn severity in the North American boreal forest region: an overview and summary of results. International Journal of Wildland Fire, 17, 443–462.
- Parmesan, C., M.D. Morecroft, Y. Trisurat, R. Adrian, G.Z. Anshari, A. Arneth, Q. Gao, P. Gonzalez, R. Harris, J. Price, N. Stevens, and G.H. Talukdarr, 2022: Terrestrial and Freshwater Ecosystems and Their Services. In: Climate Change 2022: Impacts, Adaptation and Vulnerability. Contribution of Working Group II to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change [H.-O. Pörtner, D.C. Roberts, M. Tignor, E.S. Poloczanska, K. Mintenbeck, A. Alegría, M. Craig, S. Langsdorf, S. Löschke, V. Möller, A. Okem, B. Rama (eds.)]. Cambridge University Press, Cambridge, UK and New York, NY, USA, pp. 197–377, doi:10.1017/9781009325844.004
- Rosalie, J., & Lynham, R. (2008). Remote sensing of burn severity: experience from western Canada boreal fires. International Journal of Wildland Fire, 17, 476–489.
- Sahwan, W. (n.d.). GEO-IT - The Technology of Data Acquisition for Sustainable Development and Crisis Management (Germany, Jordan, Lebanon and Syria). Freie Universität. Accessible at: https://www.geo.fu-berlin.de/en/v/geo-it/gee/2-monitoring-ndvi-nbr/2-1-basic-information/index.html
- Thapa, P. (2021). Forest Fire Area Detection using Satellite Remote Sensing Technologies.